

CHEMISTRY**PAPER 1 (THEORY)****Maximum Marks: 70****Time Allowed: Three Hours***Reading Time: Additional Fifteen Minutes***Instructions to Candidates**

1. You are allowed an additional fifteen minutes for only reading the question paper.
2. You must NOT start writing during the reading time.
3. This question paper has 8 printed pages.
4. It is divided into four sections and has twenty one questions in all.
5. Answer all questions.
6. Section A has fourteen subparts. Each question carries 1 mark.
7. While attempting Multiple Choice Questions in Section A, you are required to write only ONE option as the answer.
8. Section B has ten questions. Each question carries 2 marks.
9. Section C has seven questions. Each question carries 3 marks.
10. Section D has three questions. Each question carries 5 marks.
11. Internal choices have been provided in two questions each in Sections B, C and D.
12. The intended marks for questions are given in brackets [].
13. All working, including rough work, should be done on the same sheet as, and adjacent to the rest of the answer.
14. Balanced equations must be given wherever possible and diagrams where they are helpful.
15. When solving numerical problems, all essential workings must be shown.
16. In working out problems, use the following data:
Gas constant $R = 1.987 \text{ cal deg}^{-1} \text{ mol}^{-1} = 8.314 \text{ J K}^{-1} \text{ mol}^{-1} = 0.0821 \text{ dm}^3 \text{ atm K}^{-1} \text{ mol}^{-1}$, $1 \text{ l atm} = 1 \text{ dm}^3 \text{ atm} = 101.3 \text{ J}$.
1 faraday = 96500 coulombs, Avogadro's number = 6.023×10^{23} .

Instruction to Supervising Examiner

Kindly read aloud the Instructions given above to all the candidates present in the examination hall.

SECTION A – 14 MARKS**Question 1**

- (A) Fill in the blanks by choosing the appropriate word(s) from those given in the brackets.
[benzaldehyde, benzyl alcohol, Rosenmund, Stephen, five, three, electrochemical, chemical, +2, +3, +4, +6, picric acid, o-nitrophenol, more, less]
- (i) Benzoyl chloride is reduced by hydrogen over palladium on barium sulphate to _____; [1]
this is known as _____ reduction. (*Recall*)

- (ii) The quantity of electricity required to reduce one mole of MnO_4^- to Mn^{2+} is _____ faradays, and the mass of a substance deposited by one coulomb of electricity is called its _____ equivalent. (*Apply*) [1]
- (iii) When acidified KMnO_4 acts as an oxidising agent, manganese changes from +7 to _____, and when acidified $\text{K}_2\text{Cr}_2\text{O}_7$ acts as an oxidising agent, chromium changes from +6 to _____. (*Apply*) [1]
- (iv) Phenol on treatment with concentrated nitric acid gives 2,4,6-trinitrophenol, commonly called _____, which is a much _____ acidic compound than phenol. (*Understand*) [1]
- (B) Select and write the correct alternative from the choices given below.
- (i) Bromoethane on heating with alcoholic potassium cyanide mainly gives: (*Understand*) [1]
- ethyl isocyanide
 - propanenitrile
 - ethanol
 - ethene
- (ii) Consider the following statements about sucrose: [1]
- P: It is a non-reducing sugar.
 Q: On hydrolysis it gives equimolar amounts of D-glucose and D-fructose.
 R: Hydrolysis changes the sign of rotation from dextro to laevo.
 S: The glycosidic linkage joins C1 of α -D-glucose to C4 of β -D-fructose.
 The correct statements are:
 (*Understand*)
- P and Q only
 - P, Q and R only
 - Q, R and S only
 - P, Q, R and S
- (iii) The coordination number and the oxidation number of iron in $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$ are respectively: [1]
- (*Understand*)
- 3 and +3
 - 6 and +3
 - 6 and +2
 - 3 and +6
- (iv) For a reaction $\text{A} \rightarrow \text{P}$, the half-life doubles when the initial concentration of A is doubled. The order of the reaction is: (*Analyse*) [1]
- first
 - zero
 - second
 - one-half
- (v) Which of the following reactions gives a secondary amine? (*Understand*) [1]
- $\text{CH}_3\text{NC} + \text{LiAlH}_4$
 - $\text{CH}_3\text{CN} + \text{LiAlH}_4$
 - $\text{CH}_3\text{CONH}_2 + \text{Br}_2/\text{KOH}$
 - $\text{CH}_3\text{NO}_2 + \text{Sn}/\text{HCl}$

- (vi) Given below are two statements marked Assertion and Reason. Read the two statements carefully and select the correct option. [1]
- Assertion:** o-Nitrophenol is steam volatile whereas p-nitrophenol is not.
- Reason:** o-Nitrophenol has intramolecular hydrogen bonding whereas p-nitrophenol is associated through intermolecular hydrogen bonding.
- (Analyse)
- (a) Both Assertion and Reason are true, and Reason is the correct explanation for Assertion.
- (b) Both Assertion and Reason are true, but Reason is not the correct explanation for Assertion.
- (c) Assertion is true and Reason is false.
- (d) Both Assertion and Reason are false.
- (vii) Given below are two statements marked Assertion and Reason. Read the two statements carefully and select the correct option. [1]
- Assertion:** The limiting molar conductivity of acetic acid cannot be obtained by extrapolating its Λ_m against $\sqrt{c}$ graph to zero concentration.
- Reason:** According to Kohlrausch's law, the limiting molar conductivity of an electrolyte is the sum of the limiting molar conductivities of its cation and anion.
- (Analyse)
- (a) Both Assertion and Reason are true, and Reason is the correct explanation for Assertion.
- (b) Both Assertion and Reason are true, but Reason is not the correct explanation for Assertion.
- (c) Assertion is true and Reason is false.
- (d) Both Assertion and Reason are false.
- (C) Read the passage carefully and answer the questions that follow.
- When a solution and pure solvent are separated by a semipermeable membrane, solvent molecules pass into the solution. This is osmosis, and the extra pressure that must be applied to the solution to stop it is the osmotic pressure, $\pi = iCRT$. Solutions of equal osmotic pressure are isotonic. The fluid inside red blood cells is isotonic with a 0.9% (w/v) solution of sodium chloride, called normal saline, which is why drugs are injected intravenously in saline. In a more concentrated (hypertonic) solution the cells lose water.
- If a pressure larger than the osmotic pressure is applied on the solution side, pure solvent is forced out through the membrane. This is used in coastal cities to obtain drinking water from sea water.
- (i) What will happen to red blood cells placed in a 2% (w/v) NaCl solution? (*Understand*) [1]
- (ii) Calculate the osmotic pressure of normal saline at body temperature, 37 °C, assuming complete dissociation of NaCl. (Na = 23, Cl = 35.5) (*Apply*) [1]
- (iii) Name the process used to obtain drinking water from sea water described in the passage. (*Recall*) [1]

SECTION B – 20 MARKS

Question 2

Rahul said that a copper spoon can safely be used to stir a solution of silver nitrate. Sneha said that an iron spatula can safely be used to stir a solution of copper sulphate. (Given: $E^\circ(\text{Ag}^+/\text{Ag}) = +0.80 \text{ V}$, $E^\circ(\text{Cu}^{2+}/\text{Cu}) = +0.34 \text{ V}$, $E^\circ(\text{Fe}^{2+}/\text{Fe}) = -0.44 \text{ V}$)

- (i) Is Rahul correct? Justify. (*Evaluate*) [1]
(ii) Is Sneha correct? Justify. (*Evaluate*) [1]

Question 3

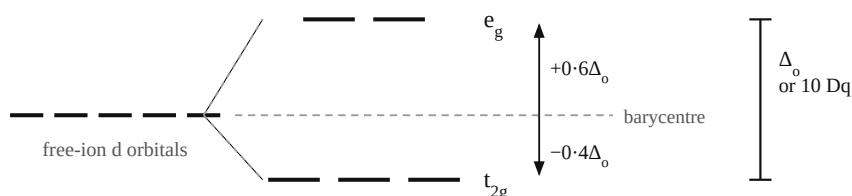
A compound [A], $\text{C}_4\text{H}_8\text{O}$, does not reduce Tollens' reagent but gives a yellow precipitate when warmed with iodine and sodium hydroxide. On reduction with NaBH_4 , [A] gives [B], which is optically active. Identify [A] and [B] and write the iodoform reaction of [A]. (*Understand*) [2]

Question 4

- (i) Answer the following with respect to the complex ion $[\text{Fe}(\text{CN})_6]^{4-}$. (Atomic number of Fe = 26)
(a) What is the hybridisation of the central metal ion? (*Understand*) [1]
(b) Comment on the magnetic nature of the complex ion. (*Analyse*) [1]

OR

- (ii) Observe the diagram below of splitting of d orbitals in an octahedral field and answer the questions that follow.



- (a) Write the electronic configuration of a d^4 ion if $\Delta_o < P$, and if $\Delta_o > P$. (*Understand*) [1]
(b) Calculate the crystal field stabilisation energy of the high-spin d^4 ion. (*Analyse*) [1]

Question 5

A piece of wood from an ancient boat shows 80% of the carbon-14 activity found in fresh wood. Estimate the age of the boat. (Half-life of carbon-14 = 5730 years) (*Apply*) [2]

Question 6

How can the following conversions be carried out? (Write the chemical equations.)

- (i) Aniline to fluorobenzene (*Recall*) [1]
(ii) Propene to propan-1-ol (*Recall*) [1]

Question 7

Substantiate the following statements with reasons.

- (i) Among the bivalent ions of the first transition series, Mn^{2+} shows the maximum paramagnetism. (Atomic number of Mn = 25) (*Understand*) [1]
- (ii) Transition elements show variable oxidation states. (*Understand*) [1]

Question 8

Answer the following:

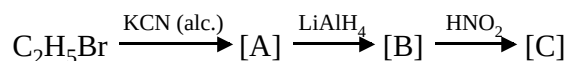
- (i) p-Cresol (4-methylphenol) is a weaker acid than phenol. Explain. (*Analyse*) [1]
- (ii) What happens when ethanol is heated with concentrated sulphuric acid at 443 K? Write the equation. (*Recall*) [1]

Question 9

- (i) Illustrate the following reactions with an example in each case:
- (a) Diazotisation (*Recall*) [1]
- (b) Coupling reaction (*Recall*) [1]

OR

- (ii) Study the reaction sequence below. [B] on warming with CHCl_3 and alcoholic KOH gives a foul-smelling compound [D]. Identify [A], [B], [C] and [D]. [2]



(*Understand*)

Question 10

The rate constant of a reaction is $4.0 \times 10^{-3} \text{ s}^{-1}$ at 300 K and $1.6 \times 10^{-2} \text{ s}^{-1}$ at 320 K. Calculate the activation energy of the reaction. (*Apply*) [2]

Question 11

Acetic acid dimerises in benzene. A solution of 0.20 g of acetic acid in 20 g of benzene lowers the freezing point of benzene by 0.45 K. Calculate the percentage association of acetic acid. (K_f of benzene = $5.12 \text{ K kg mol}^{-1}$, molar mass of acetic acid = 60 g mol^{-1}) (*Apply*) [2]

SECTION C – 21 MARKS

Question 12

- (i) Of bromoethane and 2-bromo-2-methylpropane, which reacts faster with aqueous NaOH by the $\text{S}_{\text{N}}2$ mechanism? Give a reason. (*Analyse*) [1]
- (ii) Write the equation for the reaction of chlorobenzene with chlorine in the presence of anhydrous FeCl_3 and name the products. (*Understand*) [1]
- (iii) What happens when iodoform is heated with silver powder? Write the equation. (*Recall*) [1]

Question 13

- (i) In a first-order reaction the concentration of the reactant falls from 0.80 mol L^{-1} to 0.25 mol L^{-1} in 20 minutes. Calculate the rate constant and the half-life of the reaction. (*Apply*) [2]
- (ii) What is a pseudo first-order reaction? Give one example. (*Understand*) [1]

Question 14

- (i) A student tried to prepare 2-methoxy-2-methylpropane by heating sodium methoxide with 2-chloro-2-methylpropane, but obtained an alkene instead.
- (a) Explain why, with the equation for the reaction that took place. (*Analyse*) [1]
- (b) Write the correct pair of reactants for this Williamson synthesis, with the equation. (*Apply*) [1]
- (ii) Anisole on heating with HI gives phenol and methyl iodide, not iodobenzene. Why? (*Understand*) [1]

OR

- (iii) How will the following be converted? (Give chemical equations.)
- (a) Cumene to phenol (*Recall*) [1]
- (b) Phenol to 2,4,6-tribromophenol (*Recall*) [1]
- (c) Ethanol to ethanal (*Recall*) [1]

Question 15

- (i) Complete and balance the following chemical equation: [1]
- $$\text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{SO}_4 + \text{H}_2\text{S} \rightarrow \text{---} + \text{---} + \text{---} + \text{---}$$
- (*Understand*)
- (ii) The actinoid contraction is greater from element to element than the lanthanoid contraction. Explain. (*Analyse*) [1]
- (iii) Transition metals and their compounds act as good catalysts. Give two reasons. (*Understand*) [1]

Question 16

- (i) Name the products of hydrolysis of maltose and the linkage present in it. (*Recall*) [1]
- (ii) Distinguish between fibrous and globular proteins, giving one example of each. (*Understand*) [1]
- (iii) Why must vitamin C be supplied regularly in the diet? (*Understand*) [1]

Question 17

- (i) A solution of 0.50 g of KCl in 100 g of water freezes at -0.24°C . Calculate the van't Hoff factor and the degree of dissociation of KCl. (K_f of water = $1.86\text{ K kg mol}^{-1}$, $K = 39$, $\text{Cl} = 35.5$) (*Apply*) [3]

OR

- (ii)
- (a) An aqueous solution of urea boils at 100.13°C . Calculate its freezing point. ($K_b = 0.52$ and $K_f = 1.86\text{ K kg mol}^{-1}$ for water) (*Apply*) [2]
- (b) Why is molality preferred to molarity in expressing the concentration of a solution used for colligative-property measurements? (*Understand*) [1]

Question 18

Write the chemical equations for the following conversions: (*Recall*)

- (i) Aniline to acetanilide [1]
- (ii) Benzene to aniline [1]
- (iii) Aniline to benzonitrile [1]

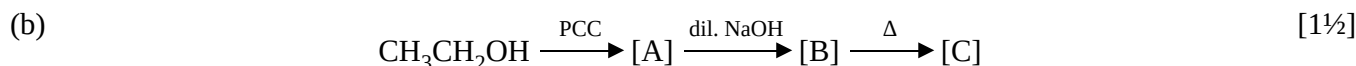
SECTION D – 15 MARKS

Question 19

- (i) An organic compound [A], $C_3H_6O_2$, liberates CO_2 with sodium hydrogencarbonate. [A] reacts with chlorine in the presence of red phosphorus to give [B], $C_3H_5ClO_2$. [A] is reduced by $LiAlH_4$ to [C], which on oxidation with PCC gives [D], C_3H_6O . [D] reduces Tollens' reagent.
- (a) Identify the compounds [A], [B], [C] and [D]. (*Understand*) [2]
- (b) Write the reaction for the formation of [B] from [A]. What is it called? (*Recall*) [1]
- (c) Which is the stronger acid, [A] or [B]? Why? (*Evaluate*) [1]
- (d) Give a chemical test to distinguish [D] from propanone. (*Analyse*) [1]

OR

- (ii) Identify the compounds [A], [B] and [C] in the following reactions: (*Analyse*)



- (iii) Write chemical equations to convert the following: (*Recall*)

- (a) Ethanal to propanone [1]
- (b) Methanal to urotropine [1]

Question 20

- (i) For the complex ion $[Pt(en)_2Cl_2]^{2+}$:
- (a) Write its IUPAC name. (*Recall*) [1]
- (b) Draw the structures of its geometrical isomers and state which one shows optical isomerism. (*Understand*) [2]
- (ii) Name the type of isomerism shown by $[Co(NH_3)_6][Cr(CN)_6]$ and $[Cr(NH_3)_6][Co(CN)_6]$. (*Analyse*) [1]
- (iii) A complex of composition $PtCl_4 \cdot 4NH_3$ gives three ions in solution and precipitates two moles of $AgCl$ per mole with $AgNO_3$. Write its structural formula and IUPAC name. (*Evaluate*) [1]

Question 21

- (i) Consider the cell



Given: $E^\circ(Cr^{3+}/Cr) = -0.74 V$, $E^\circ(Fe^{2+}/Fe) = -0.44 V$

- (a) Write the cell reaction and calculate the emf of the cell at 298 K. (*Apply*) [2]
- (b) Calculate ΔG for the cell reaction under these conditions. (*Apply*) [1]
- (c) Draw a labelled diagram of the cell and show the direction of electron flow. (*Understand*) [1]

- (d) State two functions of the salt bridge. (*Understand*) [1]

OR

- (ii)
- (a) The resistance of a conductivity cell filled with 0.1 M KCl solution (conductivity 0.0129 S cm^{-1}) is 200Ω . The same cell filled with 0.02 M NaCl solution shows a resistance of 1100Ω . Calculate the cell constant, and the conductivity and molar conductivity of the NaCl solution. (*Apply*) [3]
- (b) How long must a current of 3.0 A be passed through a solution of CuSO_4 to deposit 1.905 g of copper? ($\text{Cu} = 63.5$) (*Apply*) [2]

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